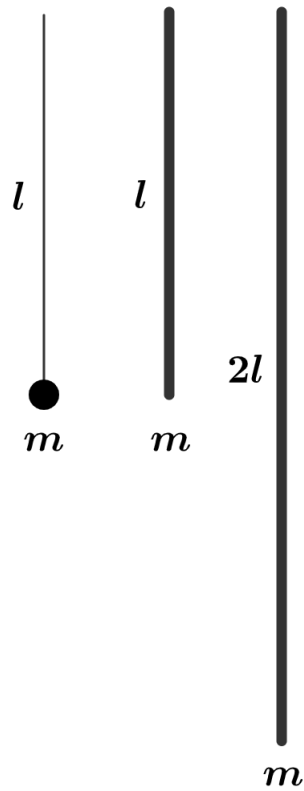


2023 F=ma Exam: Problem 17

Kevin S. Huang



Recall the moments of inertia,

$$I_{\text{point}} = mr^2$$

and

$$I_{\text{rod, end}} = \frac{1}{3}ml^2$$

In our case,

$$I_1 = ml^2$$

$$I_2 = \frac{1}{3}ml^2$$

$$I_3 = \frac{1}{3}m(2l)^2 = \frac{4}{3}ml^2$$

Thus, $I_3 > I_1 > I_2$ so the answer is E.